Final Projects

EE330 – Integrated Electronics

Iowa State University

Fall 2025

Project Policies

Project Option

This document provides a list of final project options for EE330 students. All final projects require the use of Cadence. These projects focus on either digital, analog, or mixed-signal design. Each project is expected to be completed by a two-person team, although if any student prefers to work alone, individual projects are acceptable. If help is needed in finding a partner, please note when signing up for a project.

Project Assignment

A sign-up link will be provided. This link will ask for the name of both partners if a partner has been identified. Only one of the two partners should sign up in the link; partners must belong to the same section. The sign-up link will allow each group to specify their project preferences. Attempts will be made to align project assignments with group preferences, though this alignment cannot be guaranteed. Individual team members need not be from the same laboratory section. We will restrict the project selection so that most three teams are working on any one of the projects. If there are more requests for specific projects, assignments will be handled on a first-come, first-serve basis. To help ensure a group gets the project they want, students are encouraged to sign up as soon as possible after the signup link is opened. Once the project is assigned, the team will be asked to provide an outline of the project within 1 week. It should include a description of the approach, the block diagram of the circuit that will be designed, and design specs. The team will be expected to discuss the outline with the TA that will be coordinating the project by the end of the first week.

Students who do not choose a project will be assigned to a project and students who do not choose a partner will be assigned a partner if requested. Please read each project description carefully before making a project selection.

Project Expectations

Each project will have four project expectations. These are

Demonstrated Project Checkpoints
Complete Design (schematic and layout)
Project Presentation
Project Report

Complete Design

All designs should be done in the ON 0.18μm or 45nm CMOS process. The complete design should include a layout of the circuit, including insertion of the project into a pad frame. Though all students worked on the design of a pad frame in a laboratory experiment, for uniformity, the TAs will provide the pads that can be used to form your pad frame. Post layout simulation results should be used to confirm that the design is complete. Included in your discussions with the TA what post layout simulations you will provide. A major portion of your project grade will be assigned to the successful completion of the design.

Project Presentation

A project presentation should be made to the TA after the design is complete. Schedule 30 minutes for the project presentation. The first 15 minutes will be a presentation of the design using power point slides. The remaining time will be used to answer questions about your project. These presentations should be scheduled during dead week. The TAs will provide a list of times for the project presentations.

Final Project Report

Students will be asked to submit a final project report which goes through project design. At a minimum, the report should discuss what the design does, how it works, and prove that the design functions with respect to project specifications and constraints. Plots, images, and math should be included as necessary. Treat this project report as one which would be submitted to the boss's boss; it is assumed they have a basic knowledge of what is being done but do not understand any of the specific design details. The goal of the presentation is to inform the boss of what the project consists of and whether it is functional and practical (to a degree).

General Information

Operational Amplifier

Some projects require, or are greatly helped by, the use of operational amplifiers. Internally compensated op-amp design is beyond the scope of this course but is discussed in EE435 and EE501. So instead, projects which need op-amps will be using externally compensated op-amps. (Do not confuse external op amps with externally compensated op amps. A document providing guidance on designing an externally compensated op-amp has been posted on the class's WEB site.

Comparator

Some projects require, or are greatly helped by, the use of comparators. The design of comparators is not discussed in this course. A document providing guidance on designing a comparator has been posted on the class's WEB site.

Verilog

Digital and mixed-signal projects will require a degree in Hardware Description Language (HDL) knowledge to complete. It is expected that students are proactive in learning to use an HDL (likely Verilog will be the HDL of choice) in order to complete a project, where necessary.

Projects

Temperature Sensor

Temperature sensors have a wide range of applications. This project will focus on the design of an on-chip temperature sensor that provides an analog output that varies linearly with absolute temperature (PTAT). It should operate over the range of 0C to 50C with an output of 0V at 0C and an output of 1V at 50oC. A pair of diodes will be used as a temperature sensor. The project will also likely be easier to complete if an operational amplifier is used. Since the topic of designing operational amplifiers is beyond the scope of this course, a method will be provided for designing an externally compensated operational amplifier for this project. Though we have not discussed temperature sensors in class, a 1-hour discussion with the course instructor or the TA should be sufficient to obtain the understanding needed to build this temperature sensor

Digital Potentiometer / Programmable Amplifier

This project is for the design of a digital potentiometer / amplifier / DAC integrated circuit. The digital potentiometer is similar in principle to the Maxim DS1666, but with a reduced number of taps, parallel rather than serial control of the tap position, and with a linear taper rather than audio tapes. The design should include layout and post-layout simulations results. The details of the device to be created are as follows:

The IC will provide the user with 4 functionalities. These are: a programmable non-inverting amplifier, a programmable inverting amplifier, a programmable digital potentiometer, and a DAC. For this project, an op-amp will be needed. The operating voltages for the design should be V_{DD} = 2.5V and V_{SS} = -2.5V. There should be one output pin, which will output one of the four functionalities mentioned based on a two-bit mode control signal A_0A_1. Once an option is selected, the output should change appropriately. In addition to the two-bit mode control signal, a four-bit program control input $C_0C_1C_2C_3$ should be provided to control the programmable options. Both the mode control and program control signals should use 2.5V as logic high and -2.5V as logic low. The user inputs should be expected to be both DC and AC signals, so bidirectionality should be integrated in the design. The four functions defined by the two-but mode control signal are described below:

A_0A_1 = 00

The circuit is to perform as a programmable, non-inverting, finite-gain amplifier. The user can provide an input signal along with the four-bit program control input. The output should be an amplified version of the input. The digital potentiometer (discussed below) should be used to control the gain of the amplifier in 16 steps.

A_0A_1 = 01

The circuit is to perform as a programmable, inverting, and finite gain amplifier. It is functionally the same as option A_0A_1 = 00, except the output should be inverted.

A_0A_1 = 10

The circuit is performed as a digital potentiometer. The digital potentiometer should have 16 taps with a nominal impedance between any two adjacent taps of 500Ω. The user should be able to provide a voltage on the upper potentiometer terminal, denoted as $V_{TO\,P,}$ anywhere between -2.5V and +2.5V, and a voltage on the bottom terminal, denoted as V_{BOT}, anywhere between -2.5V and V_{TOP}.

A_0A_1 = 11

The circuit is to perform as a 4-bit buffered output string DAC. The op-amp should be configured as a unity gain buffer in this application. The reference voltage for this DAC, denoted as V_{REF}, should be 1V relative to ground. (The basic digital potentiometer can be configured to form the resistor string for this DAC)

Torch Light

The term “Torch Light” refers to a flashlight that offers a high-intensity output, generally provided by a high-intensity LED. Some of the most practical Torch Lights are powered by a single rechargeable lithium-ion battery.

For this project, we designed an IC that can be used to build a torch light. The torch light should be powered by a single 18650 Lithium Ion Battery, and the light source should be a CREE XM-L2 LED. The Torch Light should have 4 user-selected modes of operation.

Mode 1: Constant output at maximum current rating of the LED

Mode 2: Flashing continuously with 50% duty cycle with approximately 2 second spacing between the flash outputs.

Mode 3: Constant output at low intensity (defined to be average power to be approximately 10% of that in Mode 1)

Mode 4: Morris Code Mode. Outputs pulse sequence of SOS in Morris Code.

There should be 2 switches. One is a Power On switch (SPST) and one is a mode select (SPSTM).

At Power On, it should enter Mode 1. Every time the mode select switch is pushed, the Mode should be advanced by 1.

Class D Audio Amplifier

This project is for the design of a class-D audio amplifier that can drive an 8Ω speaker. The circuit should have 6 pins, VDD, Gnd, Vin+, Vin- ,Vout+, Vout-. The speaker should be connected directly across the output port pins. The switches in the class-D amplifier should be sized so that the power dissipation for a 500mW speaker output is less than 10mW.

Digital-Twin Implementation of an Electronic Theremin

For the past several years there has been a growing mandate to follow establish a digital twin approach to the design of many systems with the goal of improving designer productivity and reducing design costs. In the context of this project, we will define a digital twin as follows:

A digital twin is a digital model of a physical device that serves as a digital counterpart for purposes of simulation, integration, testing, and maintenance.

This project will focus on using the digital twin approach to design an integrated electronic theremin-like sound device, henceforth referred to simply as a theremin. This has properties like a musical device that was patented in 1928, but the operation and sound that will be generated by this project is entirely different.

The physical device will be a breadboard implementation of the theremin. The digital twin will be the netlist of a chip-level theremin that will be layed out simulated.

A unique property is that it is a continuous tone rather than a discrete tone device that is "played" without any physical contact with the device.

For this project there will be 4 external photo resistors placed in an appropriate structure so that the light striking the photo resistors can be modulated by moving one's hands above the photo- sensitive device so that any one or all of the photo-sensitive devices can go completely dark by some positioning of the player's hands. Ambient light in a room should provide a light source. You can specify whatever commercially available photo-sensitive devices you want to use, but likely they will be a photo-resistor, a phototransistor, or a photodiode.

This Theremin is to be comprised of two independent but identical audio frequency oscillators with the 4 photo-sensitive devices being used to control the amplitude and frequency of the two oscillators. The output of the two oscillators is to be added together and that comprises the output of the device. The output should be capable of driving a 1K load off chip. When the instrument is played by moving hands above the photodetectors, unique sounds will be created by modulating the light striking the photo detectors and thereby controlling the amplitude and frequency of the two oscillators. With total darkness, the audio output should vanish. You may do either an analog or digital implementation. Experimental verification of the functionality of the Theremin should be provided with the digital twin being expected to provide the same functionality.

Though not required, variants of this musical device can be included by selecting different modes of operation, and the modes you may consider are limited only by your own imagination. One modification could involve using one of the oscillators to modulate the

amplitude of the second modulator. Another could involve using an input audio signal (such as that obtained from playing a .mp4 music file) to control the amplitude of one or

Both oscillators with the photo-sensitive device are being used to control the magnitude of this input signal.

Design of Perimeter

Ophthalmologists often test the field of vision of patients to help diagnose potential eye problems of a patient. Perimeters are often used for these tests. With a perimeter, light pulses or test images are presented to the patient at various angles in a three-dimensional space to determine how a patient sees at both central locations and at peripheral locations. There are both manual and automated perimeters, and the number of locations and positions may vary from one commercial product to another.

This project will be to design a 64-pixel perimeter. The design is to be completed through post-layout simulation. At each pixel location, there will be an LED. The perimeter randomly and sequentially generates a light signal at each of the pixel locations. The time between generating the test signals should be between 7 sec and 20 sec and vary enough, so the patient cannot predict when the next signal will be applied. The patient is asked to click on an SPST momentary switch every time a light is observed. A successful detection is to be recorded if the SPST is clicked within 5 seconds of applying the stimulus. At the conclusion of the test, the perimeter should have in memory the number of correct detections. Occasionally, a patient may click if there is no signal present, and this will be defined as a false click. The number of false clicks should also be recorded. After the test is over, the number of successful clicks and the number of false clicks should be displayed on a 4-digit 7-segment display. You may specify any commercial 4-digit 7-segment display. The technician conducting the test will have two switches to control the perimeter. One is a toggle switch which should be used as an input to determine whether the displayed output is the number of successful clicks or the number of false clicks. The other is an SPST push button switch that will start another test. This readout should remain active after the test is completed until another test is initiated.

Note: To keep the scope of this project manageable, we are simply counting the number of successful and false clicks. Commercial perimeters will invariably keep track at which pixel location the patient was not able to correctly detect pulses and likely display this information on a 2-d graph.

Comparative Analysis of Resistor Trimming Techniques

This research-oriented project focuses on the design and evaluation of resistor trimming techniques used in analog and mixed-signal circuits. Students will investigate two distinct resistor trimming configurations (e.g., laser trimming, network-based trimming using series/parallel or shunt structures) and perform a detailed comparative analysis.

The objective of the project is to understand how different trimming approaches impact key performance metrics. Students will model, simulate, and analyze both configurations, and determine which structure provides superior performance under varying conditions. Though we have not discussed trimming in class, a 1-hour discussion with the course instructor or the TA should be sufficient to obtain the understanding needed to complete this project.

Priority Elevator Controller

Elevator systems are widely used in residential and commercial buildings to transport passengers efficiently between floors. This project focuses on the design of a digital elevator controller capable of managing multiple floor requests while maintaining efficient and realistic operation.

The controller shall be designed for a **4-floor system** and should manage requests, determine direction of travel, and control door operation. The design should include a **priority system for the top floor**, while also implementing a direction-based scheduling strategy that minimizes unnecessary movement.

The elevator should accept floor requests from users and service them according to the following operational behavior:

- The controller shall maintain a record of all active floor requests (a queue).
- When the elevator begins moving in a given direction, it shall continue traveling in that direction as long as there are pending requests ahead.
- While traveling, the elevator shall stop at any requested intermediate floors that lie in the current direction of motion.
- The elevator shall not reverse direction until all requests in the current direction have been serviced up to the current goal floor.
- The system accepts inputs asynchronously but outputs synchronously with positive clock edge
- The control system fulfills one request per clock cycle

A unique aspect of this project is the implementation of a **top-floor priority system**. If the top floor is requested, that request shall be treated as the highest-priority destination. However, while traveling toward the top floor, the elevator must still stop at any requested floors

encountered along the way in the current direction of motion. Once the top floor has been reached and all intermediate requests have been serviced, the controller may then reverse direction and begin servicing any remaining requests.

The design shall provide the following outputs:

- ∉ All input pins should be fed to the output to be driven for display
- ∉ A signal indicating the current floor of the elevator
- ∉ A signal indicating the direction of motion (up or down)
- ∉ A signal indicating the door state (open or closed)
- ∉ A signal indicating when the controller is servicing a priority request

The controller should be implemented using a combination of registers, combinational logic, and a finite state machine. The design should include clearly defined modules for request storage, direction and scheduling logic, floor tracking, and door control.

The system should initialize an idle state on reset, with the elevator located at a defined starting floor and no active requests. The design should be verified through simulation, demonstrating correct handling of multiple simultaneous requests, proper direction-based servicing, and correct prioritization of the top floor.

Students are encouraged to design the controller in a modular fashion and to clearly document the scheduling logic used to determine elevator movement.

Testing will need to be shown both in Verilog and in cadence. Design will need to be synthesized and pass DRC/LVS.

In cadence students will need to add output load driving ability for each output pin to have the ability to be hooked up to a display system